Existing Checkpoint: Training Loss by Decision Class

Status: Owner-run loss measurement completed. Date: 2026-10-15.

Question

The frozen name-first checkpoint clarified only four of ten reviewed training inputs, despite verified assistant supervision. Does its aggregate loss hide weaker learning of CLARIFY or REFUSE answers?

This is a checkpoint diagnostic, not another training experiment. Keep the checkpoint, dataset, prompt, answer formatting and 1,024-token context unchanged.

Command

From `project-baymax`, run:

```
.venv\Scripts\python scripts\report_sft_loss.py `
  --split train `
  --max-rows 1000 `
  --max-sequence-length 1024 `
  --output tmp\name-first-training-loss-by-class.json
```

The default config selects the existing two-epoch QKVO name-first model. Use `--config` to explicitly select another config, not an inference prompt override. Optional `--dry-run` checks config, prompt, dataset and adapter identities without loading a tokenizer/model or measuring loss. Use a different output for a dry run.

Measurement

Read the first 1,000 training rows in their unchanged order. Reuse the actual training tokenization and prompt masking, including truncation. Load the existing adapter read-only, disable gradients and dropout, and score one row at a time. Do not generate, repair, execute tools, or call an optimizer.

The model's built-in causal cross-entropy scores each visible assistant token against its ground-truth preceding tokens. Prompt labels are ignored (`-100`), and position zero has no preceding prediction. Visible chat-ending tokens count. Rows with no scoreable labels are reported as skipped; partially truncated answers remain included, matching the training pipeline's existing behavior.

Report ACT (both direct and multi-tool), CLARIFY and REFUSE separately:

- scored, skipped and truncated row counts;
- scored assistant-token counts;
- token-weighted mean loss, in nats;

- unweighted mean of per-row losses.

The JSON also records overall loss, per-row file lines/source refs and loss, config/dataset/adapter hashes, actual library versions, model dtype and resolved model commit when exposed by Transformers. Existing reports cannot be overwritten; outputs inside dataset/model directories are rejected.

Interpretation Limits

Higher loss means the checkpoint assigns less probability to the supplied answer tokens on average. Token weighting measures average token prediction; row weighting prevents long answers from dominating that particular average.

Neither is routing accuracy. Common JSON syntax and easy later tokens can hide a bad initial tool choice. Teacher forcing supplies correct preceding answer tokens, unlike free generation. Answer lengths, vocabulary and task mix also differ across classes. A low loss therefore does not prove reliable decisions, and a class-loss gap alone does not establish its cause or justify more epochs.

Owner-Run Result

Source: `tmp/name-first-training-loss-by-class.json` and the owner's terminal output.

Class	Scored rows	Skipped	Truncated	Scored tokens	Loss/token	Mean row loss
ACT	774	26	32	30,864	0.039647	0.035162
CLARIFY	100	0	0	2,845	0.346352	0.330206
REFUSE	100	0	0	2,905	0.299602	0.293663

ACT contributes approximately 84% of scored tokens. The checkpoint assigns lower average probability to the supplied boundary-answer tokens than to ACT-answer tokens. This does not isolate routing from question/reason composition. All boundary rows are visible to supervision, so truncation does not explain this particular class difference.

Next diagnostic: separate call-name and text losses, without changing this checkpoint or its training data.

No benchmark examples were read and no training target was changed.

Exported 2026-10-17. Tables and saved measurements are preserved. Combined PDFs use internal document navigation. Source/result links point to the GitHub branch; unpublished local data/checkpoint links are not active.